

Document Control

- System Name: Application

Executive Summary

- This BRD defines the business requirements for the Application system, focused on ingesting customer and order CSV data into Delta tables using Delta Live Tables.
- The process includes data cleansing, computation of a TotalAmount field, merging datasets to a historical ordersummary table with SCD Type 2 logic, and calculating daily aggregate customer spending.

Business Objectives

- Ingest customer and order data from defined CSV sources into Delta tables.
- Cleanse data by removing null equivalents and duplicates.
- Enrich order records with computed TotalAmount values.
- Maintain an accurate, historical view of customer-order relationships with SCD Type 2.
- Aggregate daily customer spending and store results in a dedicated table.
- Implement the pipeline in a unified, reproducible manner using Delta Live Tables.

In-Scope

- Loading customer and order data from specified filesystem sources.
- Creation and management of Delta tables: customer, order, ordersummary, customeraggregatespend.
- Cleansing, computation, joining, SCD Type 2 handling, and aggregation logic.
- Execution and orchestration using Delta Live Tables.

Out-of-Scope

- Data sources, fields, or systems not defined in requirements.
- Security, access controls, SLAs, or monitoring.
- Performance tuning, scheduling, and automation outside explicit steps.
- Reporting, analytics, or user interface delivery.

Stakeholders

- Not explicitly defined in requirements input.

Business Requirements

BR-APP-001 — Source Ingestion

- Description: Load customer and order CSV data from specified filesystem paths into respective Delta tables.

- Business Value: Enables robust, structured storage for downstream processing.
- Priority: High

BR-APP-002 — Schema Application

- Description: Apply defined schemas to customer and order data upon ingestion, ensuring columns match requirements.
- Business Value: Standardizes data for accurate processing and reporting.
- Priority: High

BR-APP-003 — Data Cleansing

- Description: Remove literal "Null"/null values and duplicate records from source tables before further processing.
- Business Value: Ensures completeness and data quality for all subsequent steps.
- Priority: High

BR-APP-004 — Computed TotalAmount

- Description: Add TotalAmount column to order table as $\text{PricePerUnit} \times \text{Qty}$.
- Business Value: Supports sales value calculations and downstream aggregation.
- Priority: High

BR-APP-005 — Create ordersummary Table

- Description: Create ordersummary Delta table by joining customer and order datasets on CustId.
- Business Value: Provides a consolidated, historical customer-order data view.
- Priority: High

BR-APP-006 — SCD Type 2 Handling

- Description: Implement Slowly Changing Dimension (Type 2) logic to track all changes to customer records in ordersummary.
- Business Value: Retains historical detail and supports accurate analysis over time.
- Priority: High

BR-APP-007 — Aggregation to customeraggregatespend

- Description: Aggregate ordersummary data by Name and Date, computing daily total spend, and store in customeraggregatespend table.
- Business Value: Delivers actionable customer spend insights for business analysis.
- Priority: High

BR-APP-008 — Pipeline Orchestration with Delta Live Tables

- Description: Execute the entire pipeline, encompassing ingestion, cleansing, computation, joining, SCD, and aggregation via Delta Live Tables steps.
- Business Value: Simplifies operations and fosters maintainable, auditable workflows.
- Priority: High

Assumptions & Constraints

- Only features, columns, and rules present in requirements are implemented.
- Implementation uses Delta Live Tables as specified.
- SCD Type 2 logic references StartDate, EndDate, and Active semantics; explicit field listing is not available.
- No additional attributes or data model elements assumed.

Success Metrics

- 100% of customer/order records loaded without residual "Null"/duplicates.
- Computed TotalAmount present for all orders.
- All customer changes tracked historically via SCD Type 2.
- Aggregated spend values align with source data and business rules.
- Pipeline completes successfully via Delta Live Tables.

Risks & Mitigations

- Ambiguity in SCD attribute field specification may affect implementation; mitigation through clarification with stakeholders.
- Absence of explicit data types could lead to schema mismatches; mitigation through schema validation during design.
- No coverage for error handling or monitoring; recommend further analysis and requirements extension.

Approval

- Approval roles, signatures, or procedures are not present in requirements input.

Appendix

Appendix A: Source Definitions

Source Name	Type	Location	Format	Notes
customerdata	CSV	/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/customerdata	CSV	Ingest to Delta table 'customer'
orderdata	CSV	/Volumes/gen_ai_poc_databrickscoe/sdlc_wizard/orderdata	CSV	Ingest to Delta table 'order'

Appendix B: Target Definitions

Layer	Table	Columns	Notes
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	Bronze	customer	CustId, Name, EmailId, Region	Loaded from
	Bronze	order	OrderId, ItemName, PricePerUnit, Qty, Date, CustId	Loaded from
	Silver	ordersummary	CustId, Name, EmailId, Region, OrderId, ItemName, PricePerUnit, Qty, Date	Joined + SCD
spend	Gold	customeraggregatespend	Name, TotalAmount, Date	Daily custom

Appendix C: Transformations

Step	Transformation	Input	Output	Logic
	Remove null/duplicates	customer, order	cleansed tables	Remove "Null" literals and duplicates
	Compute TotalAmount	order	order	TotalAmount = PricePerUnit × Qty
	Join	customer, order	ordersummary	Join on CustId
	SCD Type 2	ordersummary	ordersummary	Track customer attribute history with StartDate/EndDate/ActiveFlag
	Aggregate	ordersummary	customeraggregatespend	Sum TotalAmount by Name, Date

Appendix D: Business Rules

Rule ID	Area	Description	Source
BR-01	Computation	TotalAmount = PricePerUnit × Qty	requirements input
BR-02	Cleansing	Remove "Null"/Null values, remove duplicates	requirements input
BR-03	Join	customer to order on CustId	requirements input
BR-04	SCD Type 2	Track changes to customer records using SCD logic	requirements input
BR-05	Aggregation	Group by Name, Date, sum TotalAmount	requirements input

Appendix E: Process Flow

Step	Description	Dependency
1	Ingest customer/order CSVs to Delta tables	Source data
2	Data cleansing and computed fields	Ingested tables
3	Join and SCD Type 2 handling	Cleansed/Enhanced tables
4	Aggregate customer spend	ordersummary
5	Orchestrate using Delta Live Tables	All process steps

Appendix F: Entity Relationship Diagram (ERD)

Text-based ERD structure:

[customer]

*CustId

Name

EmailId

Region

[order]

*OrderId

ItemName

PricePerUnit

Qty

Date

+CustId

TotalAmount

[ordersummary]

*CustId

Name

EmailId

Region

OrderId

ItemName

PricePerUnit

Qty

Date

Active

StartDate

EndDate

[customeraggregatespend]

*Name

TotalAmount

Date

Relationships:

- customer 1--* order
- customer *--* ordersummary
- order *--* ordersummary
- ordersummary *--* customeraggregatespend

This ERD captures the core table entities, their primary fields, and how the Application data objects relate through the lifecycle from ingestion to aggregation.